

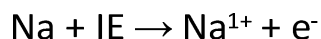
Ion Formation

Atoms are most stable when their valence shell is full, and their electron configuration is similar to that of a noble gas. In general, this means that atoms want to have 8 electrons in their valence shell. We call this a stable octet.

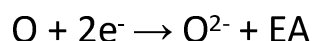
An ion is an atom that has gained or lost electrons to gain a stable octet.

- An atom that loses electrons, has a positive charge and is called a cation
- An atom that gains electrons, has a negative charge and is called a anion

- There is a third, less common ion responsible for the production of tears in mammalian tear-ducts. This ion is called the onion.



Na has 1 valence electron, so applying Ionization Energy, the electron is removed and, it achieves the same electron configuration as Ne.



O has 6 valence electron, so by gaining 2 electrons, it achieves the same electron configuration as Ne. Electron Affinity energy is released.

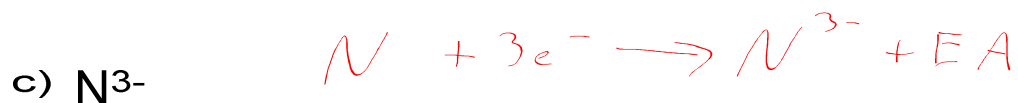
Acronym:

Negative
Electron
Affinity
Right

Positive
Ionization
Energy
Left

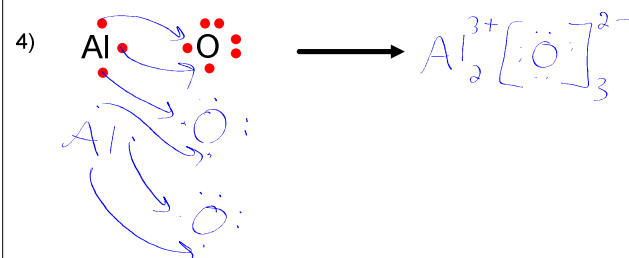
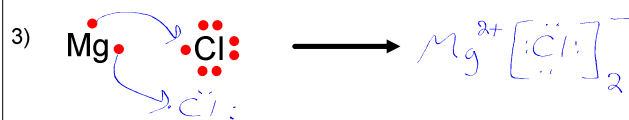
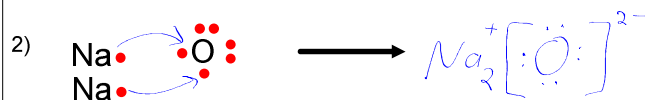
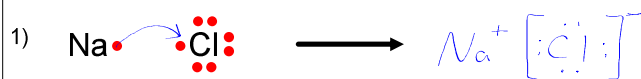
Ion Formation
Equations

Ex) Give the ion formation equations for:



Ionic Bonding occurs when atoms *exchange* electrons to satisfy their stable octets. The resulting cation(s) and anion(s) are attracted to each other by electrostatic forces and a compound is formed.

Using Lewis diagrams, we can show the formation of these ionic compounds.



Use electron dot diagrams to illustrate the formation of

- (a) lithium iodide
- (b) barium chloride
- (c) potassium oxide
- (d) calcium fluoride

Explain, referring to stable octets, how the following ionic compounds are formed from pairs of elements; illustrate the formation of each compound with ~~electron dot diagrams~~; and predict the formula of each compound.

- (a) magnesium chloride $MgCl_2$
- (b) sodium sulfide
- (c) aluminum oxide (bauxite)
- (d) barium chloride
- (e) calcium fluoride (fluorite)
- (f) sodium iodide
- (g) potassium chloride (a substitute for table salt)